XVO: Generalized Visual Odometry via Cross-Modal Self-Training

ICCV, 2023

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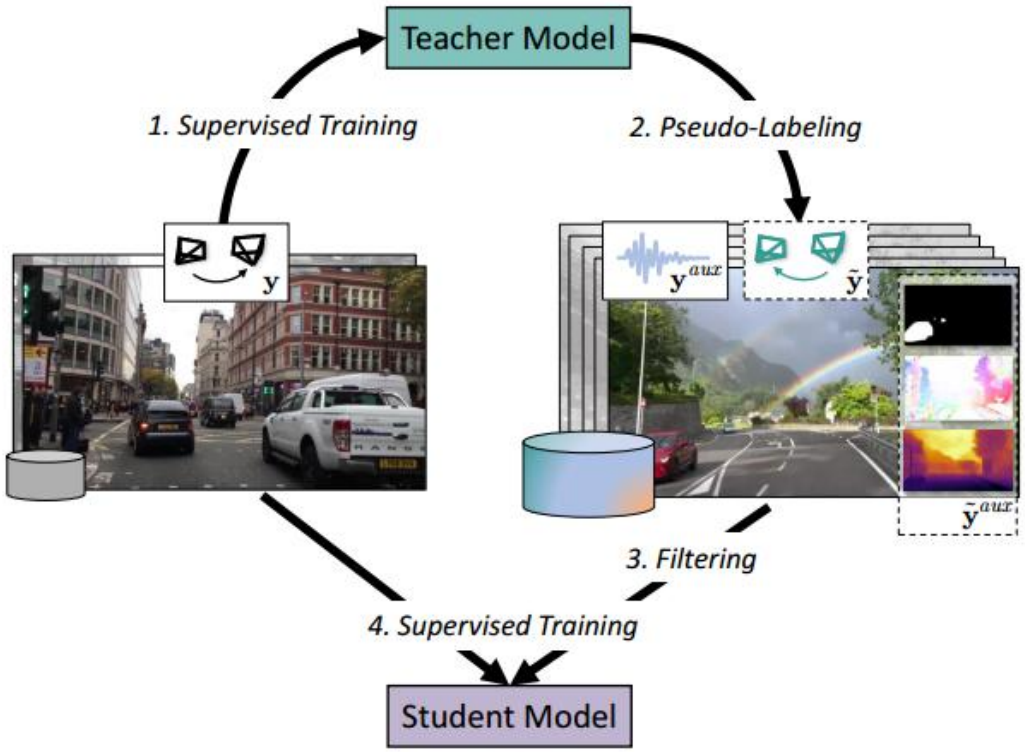
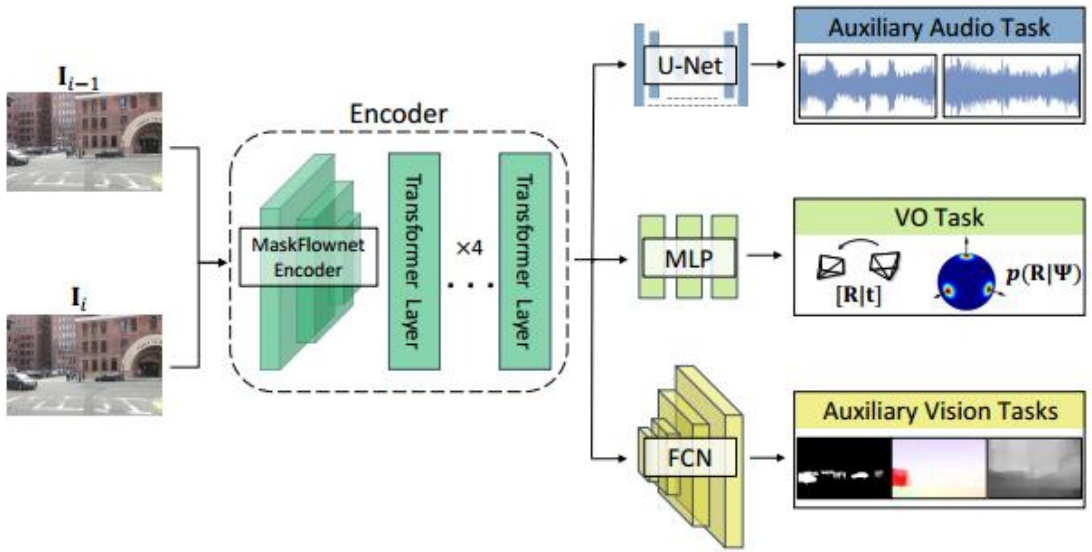
ICCV 2023	一句话概括核心思想:	引入跨模态辅助任务的自训练VO方法

问题和动机：

传统的 VO 模型通常只能在某一个特定的数据集（比如仅仅在天气晴朗的 KITTI 街道）上表现良好，并且，传统算法极其依赖预先知道相机的“内参”（焦距、光心等），有真值标签的数据极其匮乏。

为此，本文引入辅助任务以及半监督训练机制来对这些问题进行解决





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实现细节1: Pseudo-Label Selection

总结：使用教师模型对大规模的无标签YouTube数据打上伪标签，引入辅助任务，并对于伪标签进行筛选（通过信息熵并结合辅助任务）

$$H(p(\mathbf{R}|\hat{\mathbf{\Psi}})) < \tau_u$$

信息熵这个部分使用到了模型预测出平移和旋转时的不确定置信度，τ_u是一个固定的阈值，来确保网络预测足够确定被选择

实现细节2: Self-Training with Auxiliary Tasks（通过辅助任务进行自我训练）

$$\mathcal{L}_{xvo} = \mathcal{L}_{vo}(\mathbf{y}) + \lambda_u \mathcal{L}_{unc}(\mathbf{y}) + \mathcal{L}_{aux}(\mathbf{y}^{aux}, \tilde{\mathbf{y}}^{aux})$$

将损失函数设计为三个部分，其中L_{vo}是VO任务损失，是L_{unc}不确定性估计损失,L_{aux}是对于辅助任务定义的，λ_u是标量超参数

$$\mathcal{L}_{vo} = \|\mathbf{t} - \hat{\mathbf{t}}\|_2^2 + \lambda_\theta \|\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}\|_2^2$$

$$p(\mathbf{R}|\mathbf{\Psi}) = \frac{1}{c(\mathbf{\Psi})} \exp(tr(\mathbf{\Psi}^\top \mathbf{R}))$$

$$\mathcal{L}_{unc} = -\log(p(\mathbf{R}|\hat{\mathbf{\Psi}}))$$

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实现细节2: Self-Training with Auxiliary Tasks（通过辅助任务进行自我训练）

辅助任务主要分为音频、分割、深度和光流

$$\mathcal{L}_{aux} = \lambda_a \mathcal{L}_{audio} + \lambda_s \mathcal{L}_{seg} + \lambda_f \mathcal{L}_{flow} + \lambda_d \mathcal{L}_{depth}$$

对于音频解码器，包含时域（振幅）和频域（频率）两个部分

$$\mathcal{L}_{audio} = \|\mathbf{A}_i - \hat{\mathbf{A}}_i\|_2^2 + \|\mathrm{FT}(\mathbf{A}_i) - \mathrm{FT}(\hat{\mathbf{A}}_i)\|_2^2$$

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实现细节2: Self-Training with Auxiliary Tasks（通过辅助任务进行自我训练）

对于分割动态对象，动态物体引起的相对运动通常会导致姿势预测不准确，本文使用了现成模型Mask R-CNN

$$\mathcal{L}_{seg} = 1 - 2\frac{\sum_{j,k,c} \tilde{\mathbf{S}}_i \circ \hat{\mathbf{S}}_i}{\sum_{j,k,c} \tilde{\mathbf{S}}_i^2 + \sum_{j,k,c} \hat{\mathbf{S}}_i^2}$$

分子部分：算的是两张掩码图的交集面积。$\tilde{S}_i$是伪标签的分割掩码（Mask），$\hat{S}_i$是网络预测出来的掩码。
分母部分：算的是两张掩码图各自的面积之和。

对于深度和光流，损失直接表示为MSE

$$\mathcal{L}_{flow} = \|\tilde{\mathbf{F}}_i - \hat{\mathbf{F}}_i\|_2^2$$

$$\mathcal{L}_{depth} = \|\tilde{\mathbf{D}}_i - \hat{\mathbf{D}}_i\|_2^2$$

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实验结果

Method	I	S	Seq 03		Seq 04		Seq 05		Seq 06		Seq 07		Seq 10		Avg	
			t_{rel}	r_{rel}	t_{rel}	r_{rel}	t_{rel}	r_{rel}	t_{rel}	r_{rel}	t_{rel}	r_{rel}	t_{rel}	r_{rel}	t_{rel}	r_{rel}
Unsupervised Methods:																
SfMLearner [126]	✗	✗	10.78	3.92	4.49	5.24	18.67	4.10	25.88	4.80	21.33	6.65	14.33	3.30	15.91	4.67
GeoNet [112]	✗	✗	19.21	9.78	9.09	7.55	20.12	7.67	9.28	4.34	8.27	5.93	20.73	9.10	14.45	7.40
Zhan *et al.* [115]	✗	✓	15.76	10.62	3.14	2.02	4.94	2.34	5.80	2.06	6.49	3.56	12.82	3.40	8.16	4.00
UnDeepVO [45]	✗	✓	5.00	6.17	4.49	2.13	3.40	1.50	6.20	1.98	3.15	2.48	10.63	4.65	5.48	3.15
Supervised Methods:																
DeepVO [94]	✓	✓	8.49	6.89	7.19	4.97	2.62	3.61	5.42	5.82	3.91	4.60	8.11	8.83	5.96	5.79
ESP-VO [95]	✓	✓	6.72	6.46	6.33	6.08	3.35	4.93	7.24	7.29	3.52	5.02	9.77	10.2	6.16	6.66
GFS-VO [105]	✓	✓	5.44	3.32	2.91	1.30	3.27	1.62	8.50	2.74	3.37	2.25	6.32	2.33	4.97	2.26
Xue et al. [106]	✓	✓	**3.32**	2.10	2.96	1.76	2.59	1.25	4.93	1.90	**3.07**	**1.76**	3.94	1.72	3.47	**1.75**
Our Teacher (KITTI)	✓	✓	3.46	**2.00**	**1.67**	**0.70**	**2.12**	**0.92**	**3.92**	**1.46**	5.93	3.96	**3.31**	**1.52**	**3.40**	1.76
Baseline Generalized VO Methods:																
TartanVO (TartanAir) [96]	✗	✗	4.20	2.80	6.19	4.35	5.84	3.24	4.21	2.51	7.11	4.96	8.00	3.21	5.93	3.51
TartanVO (10% Noise)	✗	✗	9.33	3.12	10.88	4.71	11.77	5.39	11.88	4.52	14.70	10.74	11.76	3.61	11.72	5.35
TartanVO (20% Noise)	✗	✗	17.79	4.42	21.58	5.04	20.12	8.54	18.80	6.26	21.34	16.27	17.45	5.03	19.51	7.59
TartanVO (30% Noise)	✗	✗	25.89	7.06	34.91	4.54	22.48	10.17	19.32	5.23	19.40	13.33	25.06	8.43	24.51	8.13
Proposed Generalized VO Methods:																
Our Teacher (nuScenes)	✓	✓	26.78	4.92	26.02	2.42	23.65	8.85	23.97	6.47	30.66	20.32	20.57	6.01	25.27	8.17
Student w/o Filter	✓	✓	26.98	9.68	22.56	2.15	14.77	5.83	**11.38**	**1.62**	16.45	9.35	20.23	8.99	18.73	6.27
Student	✓	✓	20.30	3.97	16.33	1.57	11.12	4.19	15.60	5.69	7.77	3.48	19.91	5.59	15.17	4.08
XVO	✓	✓	**14.53**	**3.93**	**16.29**	**0.96**	**8.31**	**2.76**	15.31	5.49	**5.86**	**3.00**	**12.17**	**3.45**	**12.08**	**3.27**

Table 1: Analysis on the KITTI Benchmark

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实验结果

Method	KITTI 00-10			Argoverse 2			nuScenes		
	t_{err}	r_{err}	se	t_{err}	r_{err}	se	t_{err}	r_{err}	se
Baseline Generalized VO Methods:									
TartanVO w/ GT Align	6.37	3.32	/	8.55	5.77	/	9.61	6.83	/
TartanVO w/o GT Align	21.67	3.33	0.29	41.11	5.77	0.40	28.23	6.83	0.29
Proposed Generalized VO Methods:									
Teacher (nuScenes)	26.16	6.84	0.25	10.89	3.40	0.16	15.93	6.73	0.20
Student w/o Filter	20.64	5.68	0.21	10.80	7.33	0.14	9.32	4.60	0.14
Student	17.04	4.02	0.16	9.16	3.40	0.14	10.54	3.94	0.13
Student+Seg	16.31	3.77	0.16	9.17	**3.18**	0.13	11.35	4.05	0.14
Student+Flow	15.60	3.19	0.19	9.04	4.45	0.13	**9.13**	4.06	**0.13**
Student+Depth	17.49	3.89	0.20	9.25	4.11	0.13	11.86	6.46	0.15
Student+Audio	**14.37**	**3.06**	**0.16**	**8.00**	**3.08**	**0.12**	**9.26**	**3.20**	**0.12**
Student+Audio+Seg	**14.20**	**3.02**	**0.16**	8.67	3.63	0.13	11.29	3.70	0.14
Student+S+F+D	18.23	3.88	0.21	8.79	4.89	0.13	8.93	**3.44**	0.13
Student+A+S+F+D	16.74	4.40	0.18	**7.89**	3.54	**0.12**	9.98	4.36	0.15

Table 2: Average Quantitative Results across Datasets

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实验结果

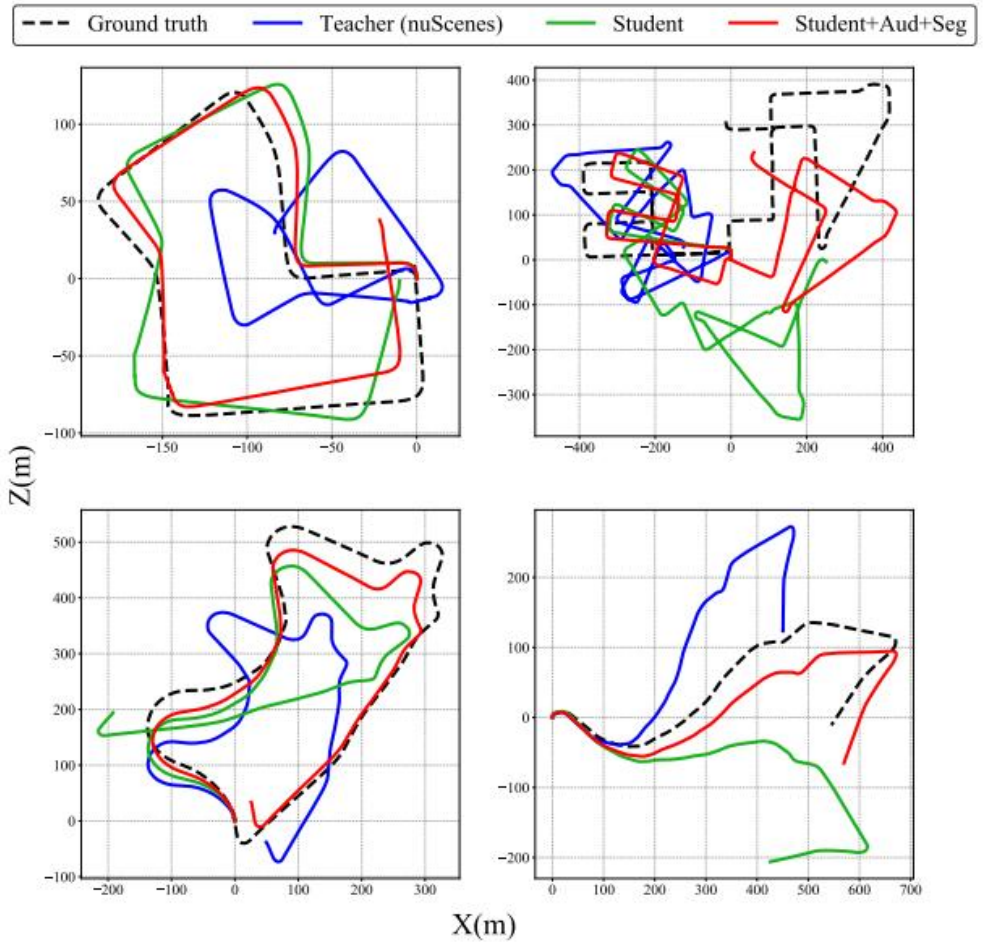

Figure 4: Qualitative Analysis on KITTI

ZeroVO: Visual Odometry with Minimal Assumptions

CVPR, 2025

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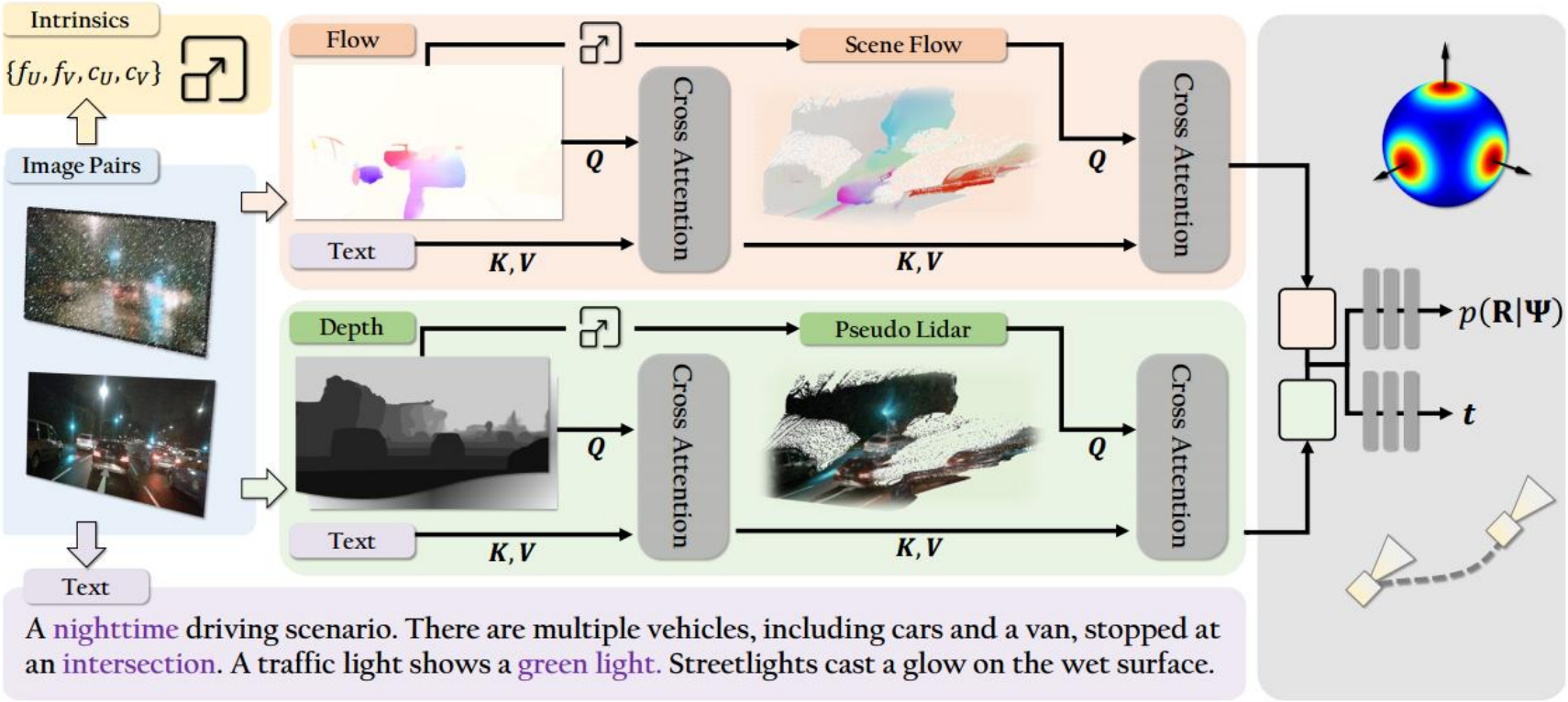
问题和动机：

本文主要为了解决三个问题，包括：

1.现有方法对于相机标定的强依赖

2.现有方法缺乏跨领域的“零样本泛化（zero-shot generalization）”能力

3.真实内在尺度（Metric Scale）的模糊性



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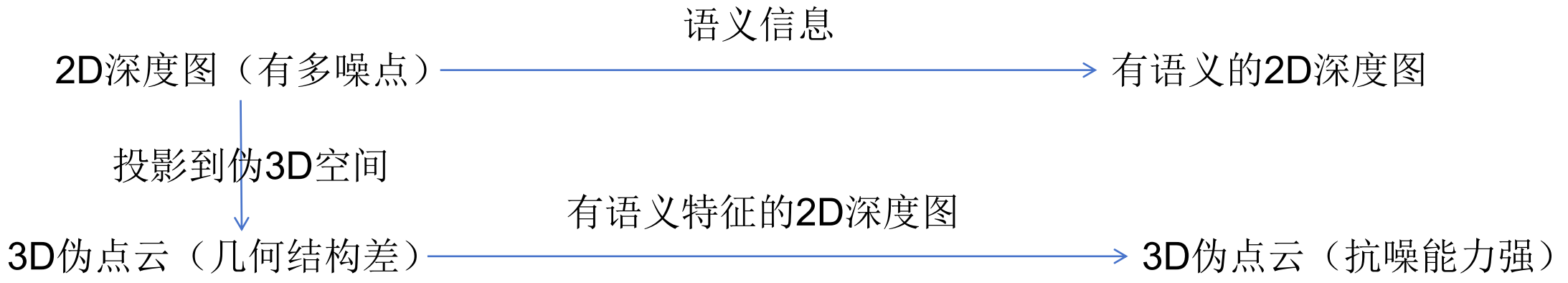
实现细节1: Language and Geometry-Guided Transformer（语言与几何引导）

融合前面提取的各种有噪声的模态数据，得出最靠谱的综合特征 。

网络首先**把2D 深度图”和“图像化的相机内参$I^{\hat{K}}$拼在一起**。然后，让文本特征 Z^l 作为引导条件，去关注和细化这些深度信息，得到中间特征 Z 。接着，再把“伪 3D 深度 $\hat{D}^{3D}$”也融合进来，得到最终的深度特征 Z^D 。对于光流，让文本特征去引导 2D 的光流对应关系F^C ，然后再融合 3D 场景流$\hat{F}^{3D}$，得到最终的光流特征 Z^F 。



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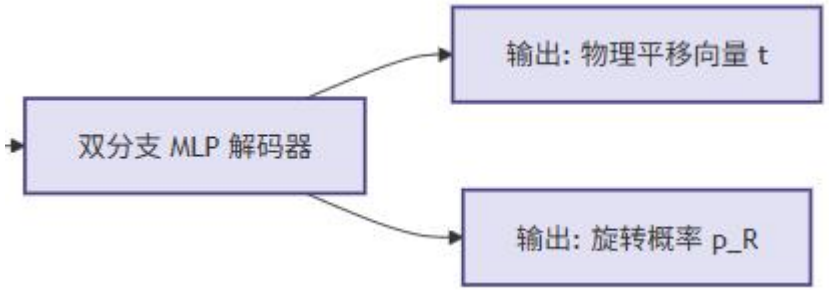
实现细节2: Probabilistic Ego-Motion Decoder（概率自运动解码器）

把 Transformer 讨论出的最终特征（Z^F和Z^D），解码成物理世界真实的相机运动位姿，并给出置信度

双分支解码器，输出平移和旋转



使用矩阵 Fisher 分布来预测旋转，并输出网络的置信度/不确定性

$$p(\mathbf{R}|\mathbf{\Psi}) = \frac{1}{c(\mathbf{\Psi})}\exp(tr(\mathbf{\Psi}^\top\mathbf{R}))$$

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实现细节3: Generalization with Semi-Supervised Training（半监督训练）

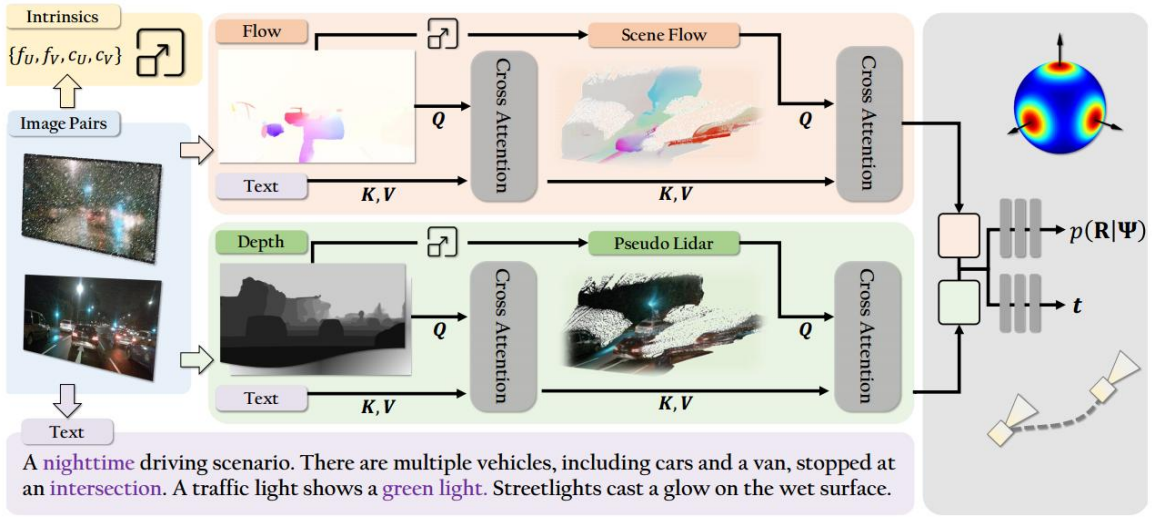
总结：可以使用野样本自己打标签（初始化），并留下好的样本进行模型训练（几何引导+语言引导）

①有监督训练--->在有标准答案（Ground Truth）的高质量数据集上进行训练，此时不依赖准确的相机内参标定。

$$\mathcal{L} = \|\mathbf{t} - \hat{\mathbf{t}}\|_2^2 - \log(p(\mathbf{R}|\hat{\mathbf{\Psi}}))$$

不需要标定得到准确的相机内参，通过transformer调低内参的权

②半监督训练--->不依赖内参，利用YouTube上的大型无约束数据集



为YouTube上的野样本打标签，留下好样本，接下来使用海量优秀标签样本进行训练

GTA数据集

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Dataset	Weathers	Lens Corruption	Resolution	Off-Road	Size	Type
KITTI [24]	Day	x	1226 × 370	x	23,201 Images	Real World
nuScenes [10]	Rain, Day, Night	✓ (rare)	1600 × 900	x	193,815 Images	Real World
Argoverse [79]	Rain, Day, Night	x	1550 × 2048	x	320,159 Images	Real World
GTA (Ours)	Snow, Rain, Day, Night	✓	1920 × 1080	✓	230,500 Images	Synthetic

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实现细节3: Generalization with Semi-Supervised Training（半监督训练）

对于伪标签的筛选有两种重要的方法：几何引导和语言引导。

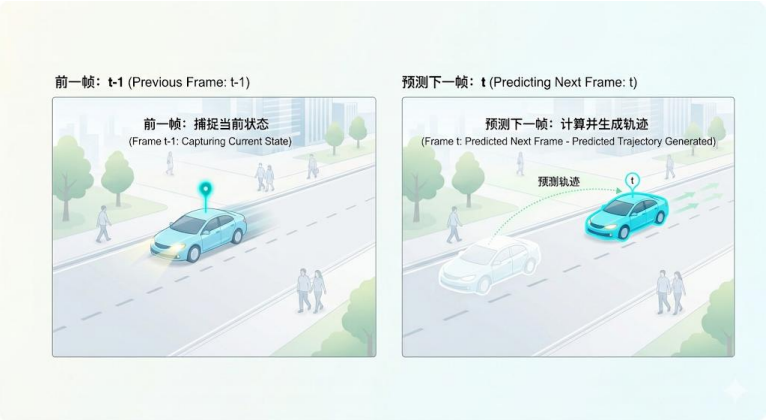
对于几何引导，预测下一帧图像的运动

$$\text{normSSIM} = \frac{\text{SSIM}(\hat{\mathbf{I}}_{i+1}, \mathbf{I}_{i+1})}{\text{SSIM}(\mathbf{I}_i, \mathbf{I}_{i+1})}$$

$SSIM(\hat{I}_{i+1}, I_{i+1})$比较预测图像与下一帧图像的相似性

$SSIM(I_i, I_{i+1})$比较前后两帧图像相似性

$$if\ normSSIM(I_i, I_{i+1}) < 0.5$$



对于语言引导，将语言特征转化成高维子空间

时间窗口$H = 10$，比较起始处和结尾处文本特征所构成的子空间

$$\text{subspace-sim} = \sin(\arccos(\text{trace}(\Lambda)))^2$$

$$\text{if subspace-sim} < 5.$$

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实验结果

Method	KITTI 00-10				nuScenes				Argoverse				GTA			
	t_{err}	r_{err}	ATE	s_{err}	t_{err}	r_{err}	ATE	s_{err}	t_{err}	r_{err}	ATE	s_{err}	t_{err}	r_{err}	ATE	s_{err}
XVO [40]	16.82	3.84	168.43	0.17	12.75	5.11	8.30	0.16	9.13	4.86	5.70	0.12	25.56	12.64	28.02	0.21
M+DS [30]	14.22	2.72	154.77	0.09	17.08	**1.46**	10.46	0.18	16.67	**1.79**	8.51	0.13	23.53	10.38	12.96	0.26
ZeroVO	7.69	2.72	105.07	0.07	10.98	4.48	6.79	0.14	6.83	3.13	4.10	0.11	14.74	10.63	8.55	0.17
ZeroVO+	**6.81**	**2.69**	**104.69**	**0.06**	**9.74**	4.37	**6.03**	**0.12**	**4.64**	2.83	**3.05**	**0.09**	**13.42**	**7.99**	**8.24**	**0.17**
LiteZeroVO+	8.85	2.90	118.54	0.08	11.57	4.44	6.87	0.13	7.65	3.82	5.28	0.11	15.93	12.16	11.26	0.18
TartanVO [76]	13.85	3.27	103.07	-	10.27	6.35	6.26	-	11.17	5.30	7.03	-	10.56	9.35	3.82	-
DPVO [69]	8.31	2.37	78.53	-	4.34	2.85	2.66	-	2.66	1.25	1.59	-	12.65	10.67	4.33	-

Table 1. Comparative Analysis Across Datasets.

$$s_{err}^{i} = 1 - \min\left(\frac{\|\hat{\mathbf{t}}_i\|_2}{\max(\|\mathbf{t}_i\|_2, \epsilon)}, \frac{\|\mathbf{t}_i\|_2}{\max(\|\hat{\mathbf{t}}_i\|_2, \epsilon)}\right), \quad s_{err} = \frac{1}{N-1}\sum_{i=2}^{N} s_{err}^{i}$$

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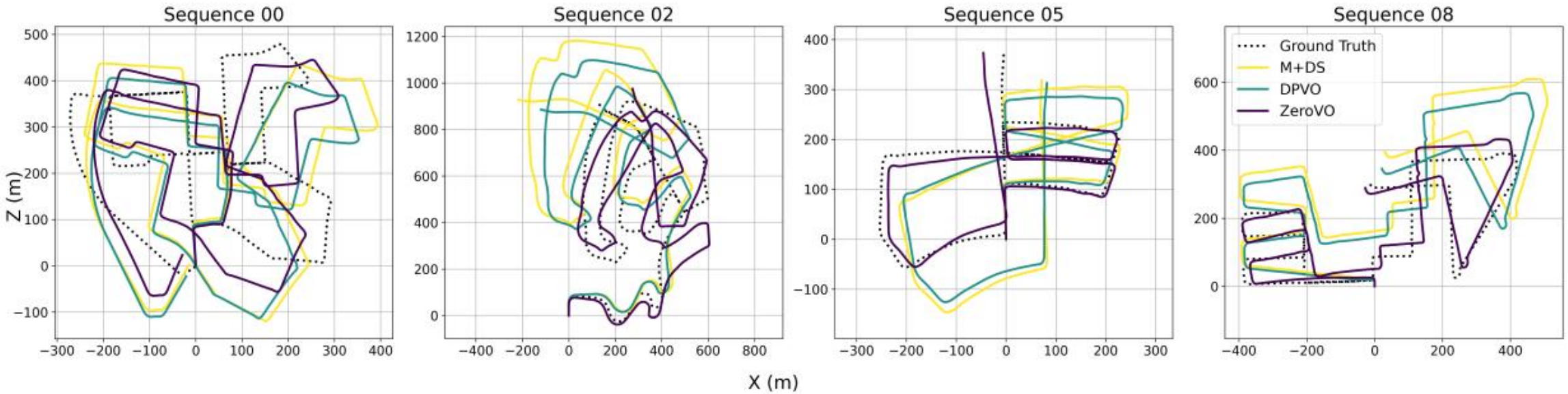
实验结果

Method	Day	Night	Rainy	Light
XVO [40]	6.61	14.41	15.99	15.73
M+DS [30]	6.08	17.19	17.49	18.54
ZeroVO	3.90	10.33	12.63	13.33
ZeroVO+	**3.60**	**10.26**	**10.10**	**11.15**

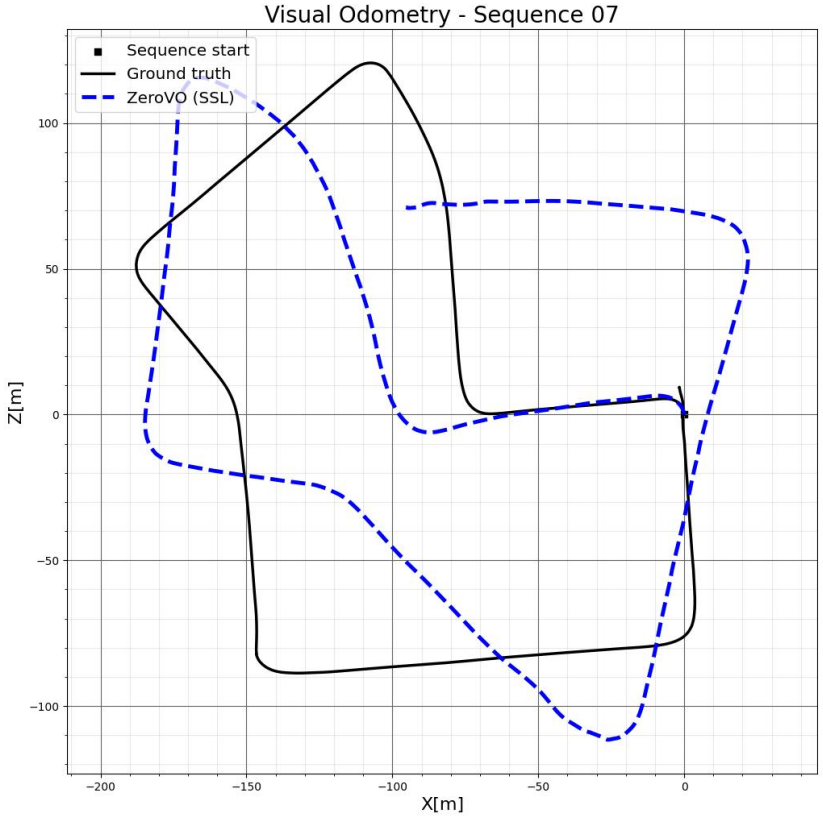
Table 3. Condition Breakdown on nuScenes



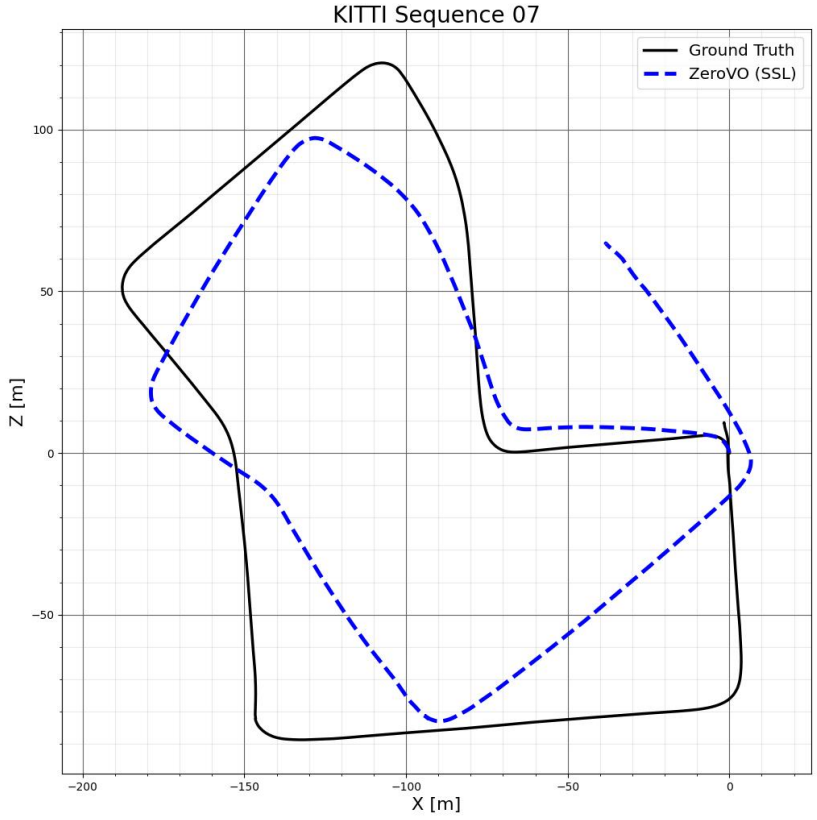
ZeroVO复现

第一阶段：进行推理

使用了KITTI数据集，使用作者提供的全监督训练权重（ZVO）和半监督训练权重（ZVO_SSL）分别进行了推理，这里只对Sequence07进行了推理



ZVO

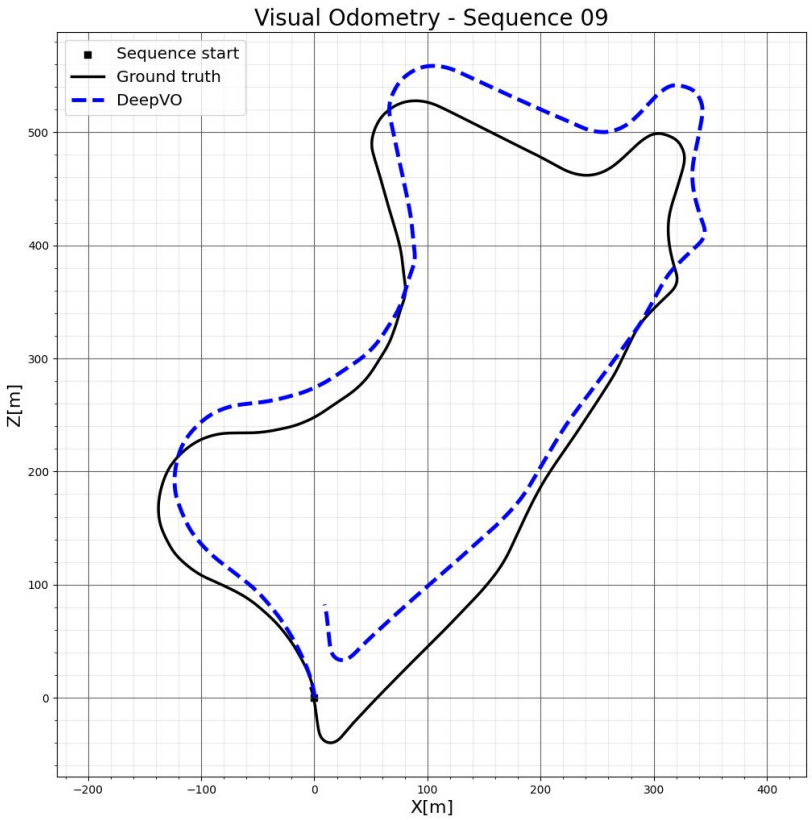


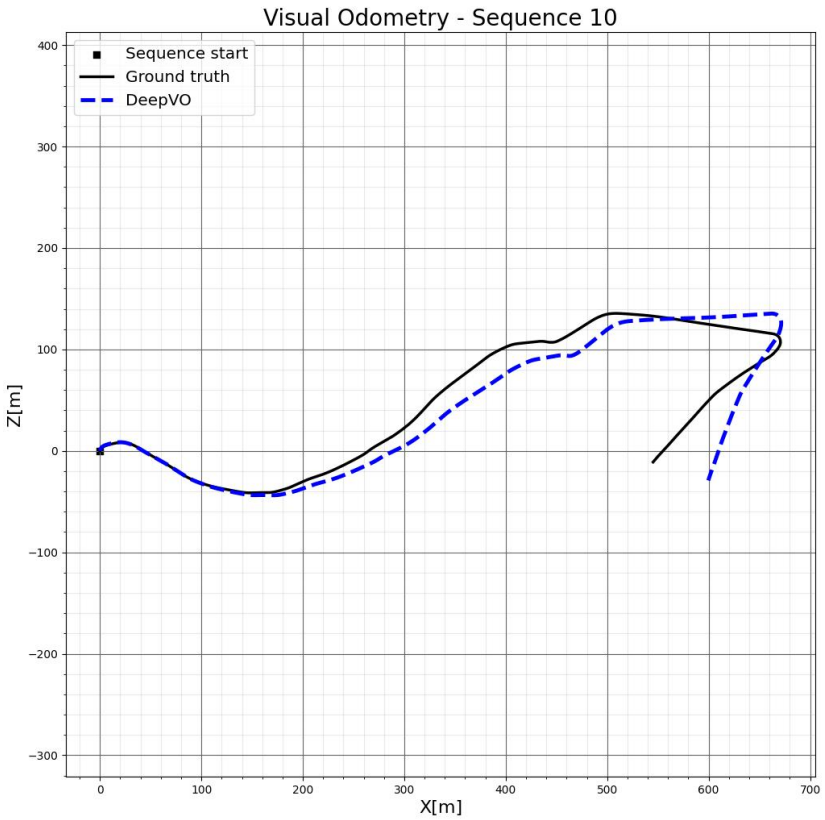
ZVO_SSL

ZeroVO复现

第二阶段：KITTI半监督训练

使用了KITTI数据集，对于00~10序列，提取深度（Metric3D），提取text（LLava），使用00~06、08进行训练，07、09、10用于测试（07用作测试是为了和推理进行对比）






```
098 NUSC_X: /, /, /, / GTA: /, /, /, / ARGO2_Stereo: /, /, /, / KITTI: 8.37, 3.47, 54.11, 0.04
099 NUSC_X: /, /, /, / GTA: /, /, /, / ARGO2_Stereo: /,_/, /, / KITTI: 8.44, 3.56, 54.38, 0.04
```

ZeroVO复现

第三阶段：RECON半监督训练

recon数据集有11836个序列，选取了前900个序列（50000多张图），提取深度（Metric3D），提取text（LLava），训练测试比例8:1

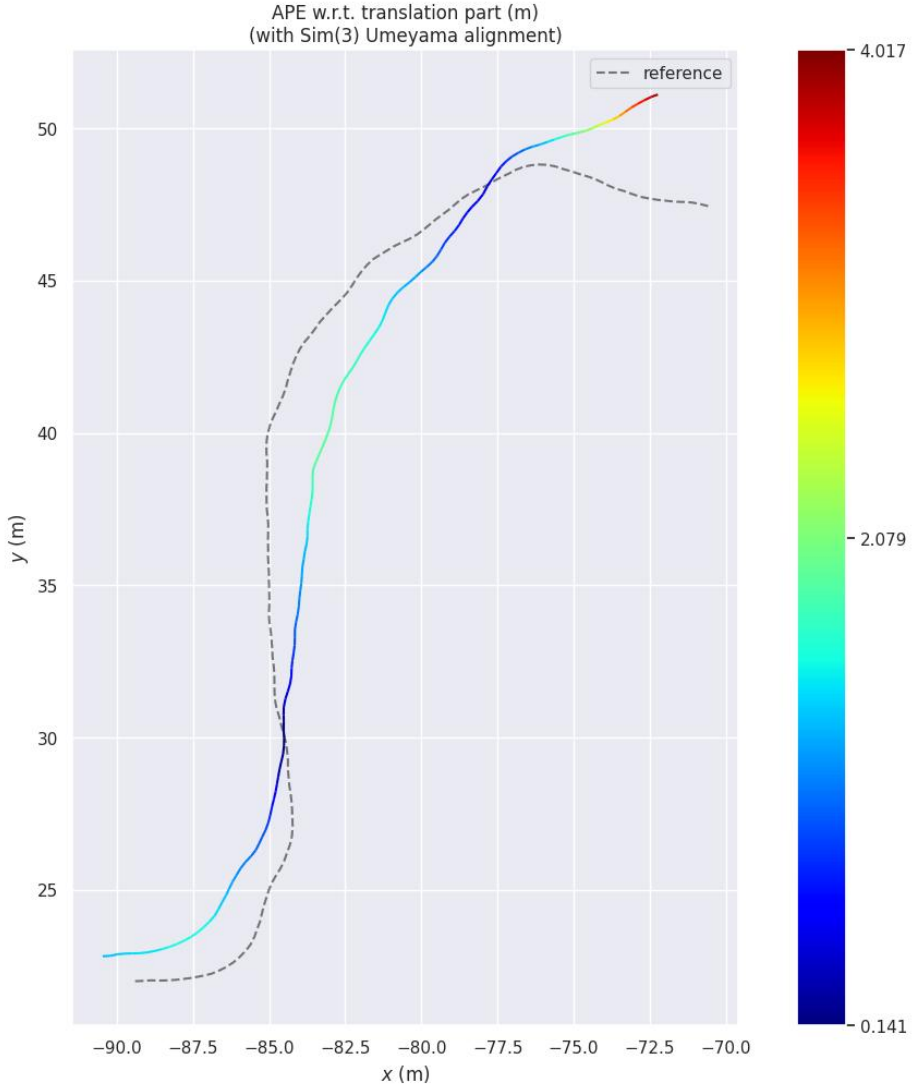
```
Evaluating RECON: 100%|                                                    | 28/28 [00:07<00:00,  3.89it/s]
Evaluating KITTI: 100%|                                                    | 28/28 [00:00<00:00, 200069.02it/s]
Evaluating NUSC_X: 100%|                                                   | 28/28 [00:00<00:00, 352674.21it/s]
Evaluating ARGO2_Stereo: 100%|                                             | 28/28 [00:00<00:00, 358050.34it/s]

==================================================
Final Statistics for ZVO_RECON_Full_800
Epoch | Scene: t_err, r_err, ate, s_err
--------------------------------------------------
000 NUSC_X: /, /, /, / RECON: /, /, 0.31, 0.00 ARGO2_Stereo: /, /, /, / KITTI: /, /, /, /
001 NUSC_X: /, /, /, / RECON: /, /, 0.28, 0.00 ARGO2_Stereo: /, /, /, / KITTI: /, /, /, /
002 NUSC_X: /, /, /, / RECON: /, /, 0.27, 0.00 ARGO2_Stereo: /, /, /, / KITTI: /, /, /, /
003 NUSC_X: /, /, /, / RECON: /, /, 0.26, 0.00 ARGO2_Stereo: /, /, /, / KITTI: /, /, /, /
004 NUSC_X: /, /, /, / RECON: /, /, 0.25, 0.00 ARGO2_Stereo: /, /, /, / KITTI: /, /, /, /
005 NUSC_X: /, /, /, / RECON: /, /, 0.27, 0.00 ARGO2_Stereo: /, /, /, / KITTI: /, /, /, /
006 NUSC_X: /, /, /, / RECON: /, /, 0.24, 0.00 ARGO2_Stereo: /, /, /, / KITTI: /, /, /, /
007 NUSC_X: /, /, /, / RECON: /, /, 0.24, 0.00 ARGO2_Stereo: /, /, /, / KITTI: /, /, /, /
008 NUSC_X: /, /, /, / RECON: /, /, 0.24, 0.00 ARGO2_Stereo: /, /, /, / KITTI: /, /, /, /
009 NUSC_X: /, /, /, / RECON: /, /, 0.24, 0.00 ARGO2_Stereo: /, /, /, / KITTI: /, /, /, /
010 NUSC_X: /, /, /, / RECON: /, /, 0.23, 0.00 ARGO2_Stereo: /, /, /, / KITTI: /, /, /, /
011 NUSC_X: /, /, /, / RECON: /, /, 0.24, 0.00 ARGO2_Stereo: /, /, /, / KITTI: /, /, /, /
012 NUSC_X: /, /, /, / RECON: /, /, 0.23, 0.00 ARGO2_Stereo: /, /, /, / KITTI: /, /, /, /
013 NUSC_X: /, /, /, / RECON: /, /, 0.24, 0.00 ARGO2_Stereo: /, /, /, / KITTI: /, /, /, /
014 NUSC_X: /, /, /, / RECON: /, /, 0.25, 0.00 ARGO2_Stereo: /, /, /, / KITTI: /, /, /, /
015 NUSC_X: /, /, /, / RECON: /, /, 0.22, 0.00 ARGO2_Stereo: /, /, /, / KITTI: /, /, /, /
016 NUSC_X: /, /, /, / RECON: /, /, 0.22, 0.00 ARGO2_Stereo: /, /, /, / KITTI: /, /, /, /
017 NUSC_X: /, /, /, / RECON: /, /, 0.23, 0.00 ARGO2_Stereo: /, /, /, / KITTI: /, /, /, /
018 NUSC_X: /, /, /, / RECON: /, /, 0.24, 0.00 ARGO2_Stereo: /, /, /, / KITTI: /, /, /, /
019 NUSC_X: /, /, /, / RECON: /, /, 0.22, 0.00 ARGO2_Stereo: /, /, /, / KITTI: /, /, /, /
020 NUSC_X: /, /, /, / RECON: /, /, 0.23, 0.00 ARGO2_Stereo: /, /, /, / KITTI: /, /, /, /
021 NUSC_X: /, /, /, / RECON: /, /, 0.24, 0.00 ARGO2_Stereo: /, /, /, / KITTI: /, /, /, /
022 NUSC_X: /, /, /, / RECON: /, /, 0.23, 0.00 ARGO2_Stereo: /, /, /, / KITTI: /, /, /, /
023 NUSC_X: /, /, /, / RECON: /, /, 0.24, 0.00 ARGO2_Stereo: /, /, /, / KITTI: /, /, /, /
024 NUSC_X: /, /, /, / RECON: /, /, 0.22, 0.00 ARGO2_Stereo: /, /, /, / KITTI: /, /, /, /
025 NUSC_X: /, /, /, / RECON: /, /, 0.25, 0.00 ARGO2_Stereo: /, /, /, / KITTI: /, /, /, /
026 NUSC_X: /, /, /, / RECON: /, /, 0.23, 0.00 ARGO2_Stereo: /, /, /, / KITTI: /, /, /, /
027 NUSC_X: /, /, /, / RECON: /, /, 0.23, 0.00 ARGO2_Stereo: /, /, /, / KITTI: /, /, /, /
```

```
--------------------------------------------------------------------------------
Loaded 62 poses from: ../data/RECON/poses/805.txt
Loaded 62 poses from: ../results/ZVO_RECON_Full_800/015/RECON/805.txt
--------------------------------------------------------------------------------
Aligning using Umeyama's method... (with scale correction)
Rotation of alignment:
[[-0.90215046 -0.4314195   0.00132526]
 [ 0.43093136 -0.90126758 -0.04488774]
 [ 0.02055986 -0.0399244   0.99899116]]
Translation of alignment:
[-7.77757023e+01  1.45715761e+01 -3.44224138e-02]
Scale correction: 1.2504095184534547
--------------------------------------------------------------------------------
Compared 62 absolute pose pairs.
Calculating APE for translation part pose relation...
--------------------------------------------------------------------------------
APE w.r.t. translation part (m)
(with Sim(3) Umeyama alignment)

       max      0.300157
      mean      0.143142
    median      0.135531
       min      0.048026
      rmse      0.155076
       sse      1.491019
       std      0.059657

--------------------------------------------------------------------------------
Plotting results...
```

ZeroVO复现

第三阶段：RECON半监督训练

问题序列807

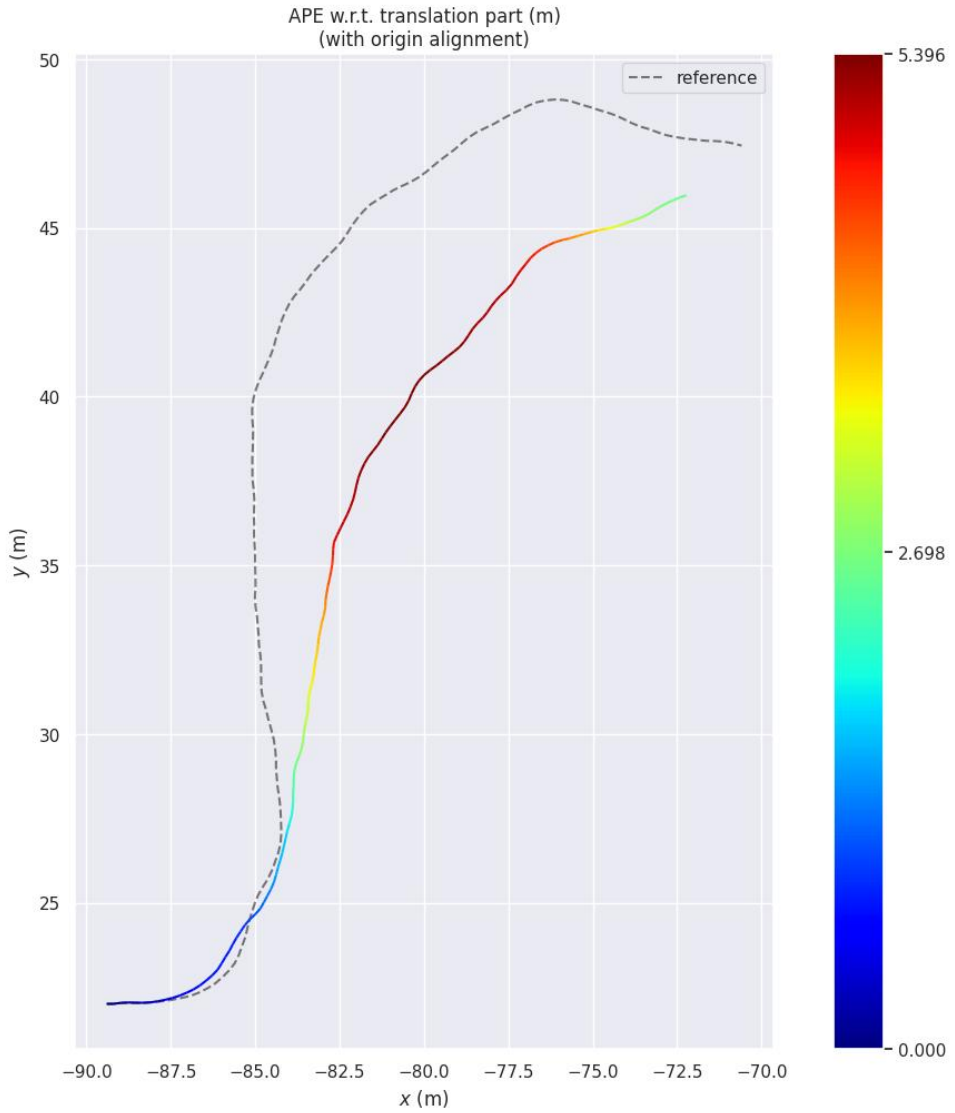


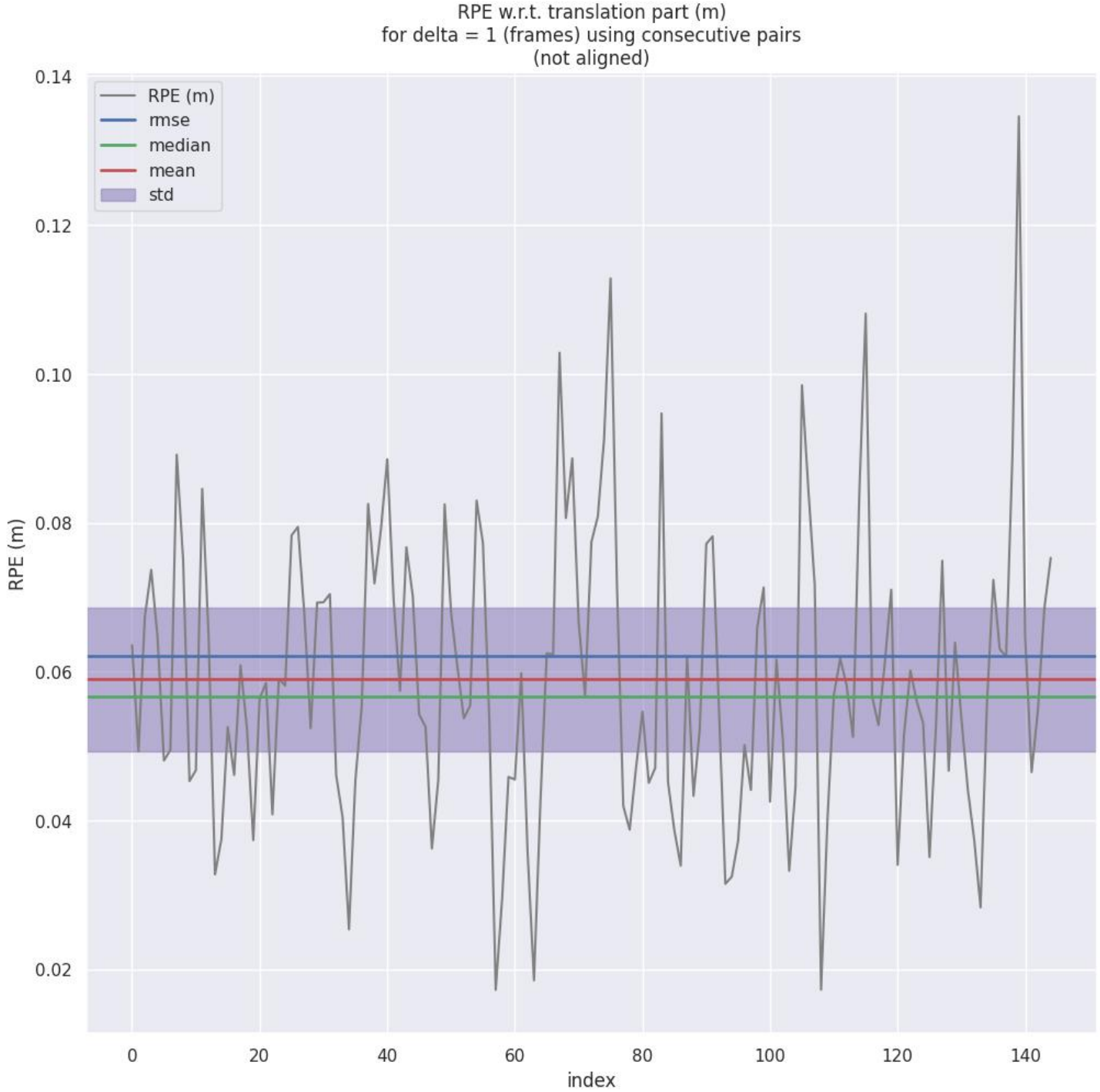

```
--------------------------------------------------------------------------------
Loaded 146 poses from: ../data/RECON/poses/807.txt
Loaded 146 poses from: ../results/ZVO_RECON_Full_800/015/RECON/807.txt
--------------------------------------------------------------------------------
Aligning using Umeyama's method... (with scale correction)
Rotation of alignment:
[[ 9.99745393e-01  2.25560254e-02 -6.12025042e-04]
 [-2.25590074e-02  9.98553034e-01 -4.88152536e-02]
 [-4.89938636e-04  4.88166315e-02  9.98807637e-01]]
Translation of alignment:
[-9.04644530e+01  2.28193247e+01  7.33200632e-02]
Scale correction: 1.1403161642809103
--------------------------------------------------------------------------------
Compared 146 absolute pose pairs.
Calculating APE for translation part pose relation...
--------------------------------------------------------------------------------
APE w.r.t. translation part (m)
(with Sim(3) Umeyama alignment)

       max      4.016575
      mean      1.355669
    median      1.344803
       min      0.141443
      rmse      1.538118
       sse      345.407684
       std      0.726615

--------------------------------------------------------------------------------
Plotting results...
```

ZeroVO复现

第三阶段：RECON半监督训练

问题序列807





ZeroVO复现

第三阶段：RECON半监督训练

问题序列807

15-18

19-22

23-26